00-D-696B
June 28, 1977
SUPERSEDING
Fed. Spec. 00-D-696a
March 11, 1964

FEDERAL SPECIFICATION

DRILLING MACHINES, RADIAL, FLOOR MOUNTED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers floor mounted, self-contained electric motor driven radial drilling machines suitable for general purpose drilling, tapping, boring, facing and reaming operations on large size or heavy workpieces.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Type and sizes. The machines covered by this specification are of the plain radial arm type and are of the following sizes. The size furnished shall be as specified (see 6.2.1):

FSC 3413

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Size A4 - 9 inch column diameter x 3 feet arm length
Size A6 - 9 inch column diameter x 4 feet arm length
Size B1 - 11 inch column diameter x 3 feet arm length
Size B2 - 11 inch column diameter x 4 feet arm length
Size B3 - 11 inch column diameter x 5 feet arm length
Size C1 - 13 inch column diameter x 4 feet arm length
Size C2 - 13 inch column diameter x 5 feet arm length
Size D1 - 15 inch column diameter x 4 feet arm length
Size D2 - 15 inch column diameter x 5 feet arm length
Size D3 - 15 inch column diameter x 6 feet arm length
Size E1 - 17 inch column diameter x 5 feet arm length
Size E2 - 17 inch column diameter x 6 feet arm length
Size E3 - 17 inch column diameter x 7 feet arm length
Size F1 - 19 inch column diameter x 6 feet arm length
Size F2 - 19 inch column diameter x 7 feet arm length
Size F3 - 19 inch column diameter x 8 feet arm length
Size G1 - 22 inch column diameter x 7 feet arm length
Size G2 - 22 inch column diameter x 8 feet arm length
Size G3 - 22 inch column diameter x 9 feet arm length
Size G4 - 22 inch column diameter x 10 feet arm length
Size H1 - 26 inch column diameter x 8 feet arm length
Size H2 - 26 inch column diameter x 9 feet arm length
Size H3 - 26 inch column diameter x 10 feet arm length
Size H4 - 26 inch column diameter x 11 feet arm length
Size H5 - 26 inch column diameter x 12 feet arm length

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Std. No. 356 - Commercial Packaging of Supplies and Equipment

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Singles copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-M-18058 - Machinery, Metal and Woodworking, Preparation for Delivery of

Military Standards:

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

National Bureau of Standards (NBS) Handbook:

H-28 - Screw-Thread Standards for Federal Services

Occupational Safety and Health Act (OSHA) of 1970:

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments	Occupational Safety and Health Administration Standards

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

American National Standards Institute (ANSI) Standards:

B5.1 - T-slots, Their Bolts, Nuts, Tongues and Cutters
B5.10 - Machine Tapers
B17.2 - Woodruff Keys and Keyseats
B92.1 - Involute Splines and Inspection

(Application for copies should be addressed to the American National Standards Institute, Dept. 969, 1430 Broadway, New York, NY 10018.)

Joint Industrial Council (JIC):

EGP - 1 JIC Electrical Standards for General Purpose Machine Tools

(Application for copies should be addressed to the National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101.)

American Gear Manufacturers Association (AGMA):

AGMA 360 - Manual for Machine Tool Gearing

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Ave., NW, Washington, DC 20005.)

National Fire Protection Association (NFPA) Standards:

NFPA No. 70 National Electrical Code
NFPA No. 79 Electrical Standards for Metalworking Machine Tools

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the product are classified herein as follows:

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3.1 Classification of requirements. The requirements for the product are classification herein as follows:

Requirement	Paragraph
First article approval	3.2
General requirements	3.3
Detail requirements	3.4
Performance requirements	3.5

3.2 First article approval. If first article approval is required (see 6.2.1 and 6.3), inspection therefor shall be in accordance with section 4. A first article comprises a preproduction model or initial production sample; approval involves testing and evaluating the first article for conformance with the requirements herein before, or in the initial stage of production under a contract.

3.3 General requirements.

3.3.1 Design. The machine shall be new and one of the manufacturer's current models. The components of the machine shall include a floor mounting base which along with workpieces, supports a column with rotating, and plain type radial arm with traversing drill head. The machine design and construction shall include components, parts and features necessary to meet the accuracy and performance requirements specified herein. All components or parts subject to wear, breakage or distortion shall be accessible for adjustment, replacement, and repair.

3.3.1.1 Safety devices. The machine shall be fitted with safety devices or covering for all parts that present safety hazards. The devices shall include guards for belts, pulleys, chains, and gears. All guards shall be removable to provide access to the guarded parts. Except as otherwise specified by the procuring activity, the machine shall comply with the standards promulgated under OSHA applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

3.3.1.2 Lubrication. All bearings (except sealed for life type), mating gears, and sliding parts shall be provided with means for lubrication. Oil reservoirs shall be fitted with oil level sight gages. Manually operated systems for ways and remote areas shall have control handles mounted in accessible locations. All oil holes, grease fittings, and filler caps shall be easily accessible.

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3.3.1.3 Interchangeability. To provide for replacement of worn parts, all parts bearing the same part number shall be functionally interchangeable, and shall be dimensionally identical within the manufacturing tolerance limits in use by their manufacturer.

3.3.1.4 Material. All parts used in the fabrication of the machine shall be new and of the quality necessary to produce a machine to meet the requirements described herein.

3.3.2 Construction. The machine shall be complete so that when installed and connected to the specified source of power it can be used for any operation for which it was designed. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.2.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where included in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.2.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing change of tightness. All such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.2.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

3.3.2.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.2.6 Threads. Unless otherwise specified (see 6.2.1), all threaded parts shall conform to Handbook H28.

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3.3.2.7 Gears. All gears used in the machine and its components shall be machined in the inch or metric system and shall conform to the applicable requirements of AGMA 360.

3.3.2.8 Plates. All words on speed and feed indicating plates and on instruction plates shall be in the English language, engraved, embossed or stamped in bold-face letters on a contrasting background.

3.3.2.9 Dials. Unless otherwise specified (see 6.2.1), all feed dials shall be graduated in the inch or metric system. Graduations to indicate tool movement shall be in increments of not more than 1/16 of an inch or the equivalent on the metric scale. Dial diameters shall be such that the graduations may be easily read. The dials shall have a satin finish chrome face or other non-reflective finish, permanently and legibly engraved, stamped or etched.

3.4 DETAIL REQUIREMENTS

3.4.1 Components.

3.4.1.1 Base. The base shall be a one-piece ferrous casting. The top or the base shall be machined for mounting the column and to provide a work surface. The work surface shall have not less than three equally spaced and parallel tee slots, running lengthwise for accepting 3/4 inch or larger tee bolts. T-slots, if machined, shall be in accordance with ANS B5.1. A coolant fluid trough shall extend around the periphery of the work surface and be connected to a coolant reservoir which shall be integral with the base. A screen to prevent chips from entering the reservoir shall be furnished.

3.4.1.1.1 Plain type table. When specified (see 6.2.1), the machine shall have a plain type table configured and machined for attachment to the work surface of the base. The plain table shall be of box construction with machined top and side working surfaces. The top surface shall have three or more tee slots and the side surface two or more tee slots. Machined tee slots shall be in accordance with ANS B5.1.

3.4.1.1.2 Universal tilting type table. When specified (see 6.2.1), the machine shall have a universal tilting type table configured and machined for attachment to the work surface of the base. The table shall consist of a base on which shall be mounted a tilting work table with two machine-finished work surfaces at 90 degrees to each other. The top surface of the table shall be not less than 20 by 20 inches.

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The side work surface shall be not less than 10 by 20 inches. The table surfaces shall be machined and provided with T-slots, three on the top and two on the side. Machined T-slots shall be in accordance with ANS B5.1. The table shall be arranged so that either work surface can be set in a vertical position by means of a worm and segment gear and have means for securing the work surfaces in position. The segment shall be graduated in degrees to indicate the angle at which the table is set. The base of the table shall be machined and provided with slots or bolt holes.

3.4.1.2 Column. The column shall consist of two ferrous castings, tubular in construction, with an inner member or trunk and an outer member or sleeve. The trunk shall be reinforced by internal ribs and shall have a flange for mounting on the base. The outer sleeve shall rotate 360 degrees around the trunk on ball or roller bearings. The column shall be arranged with a clamping mechanism for securing the trunk to the sleeve. The column sleeve shall be machined for receiving and mounting the arm. The finish on the surface giving support to the arm shall be not rougher than 16 microinches.

3.4.1.3 Arm. The arm shall be a reinforced ferrous casting, machined and fitted to the column. The length of the arm bearing on the column shall be not less than twice the diameter of the column. Means shall be provided for lubricating the column sleeve when the arm is raised and lowered. Wipers shall be provided for cleaning the column sleeve. Ways shall be provided for movement of the head. The way which provides the main support for the head shall have a surface hardness of not less than Rockwell C-50; it may be a replaceable strip way. The arm shall have a rack gear for transversing the head along the arm. The arm shall rotate through 360 degrees by means of the column sleeve rotation. Vertical arm movement shall be accomplished by a elevating screw and nut assembly driven either separately or by the main drive motor. The nut assembly shall be mounted on ball or roller bearings and shall be equipped with a safety device that supports the arm in case of thread stripping Automatic limit stops shall be provided to arrest the travel of the arm at both ends of the vertical travel range. A clamping device shall be furnished for securing the arm to the column. An interlock or other means shall be provided to prevent engagement of the elevating mechanism when the arm is clamped to the column. When specified (see 6.2.1), machines with column diameters larger than 13 inches, shall have either electric, hydraulic, or electrohydraulic clamping devices for securing the arm to the column and the column trunk to the column sleeve.

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3.4.1.4 Head. The head shall be fully enclosed, machined and fitted to the arm. A clamp shall be provided for locking the head to the arm. The head shall be traversed along the arm by means of a power or manually operated rack and pinion gear mechanism. When specified (see 6.2.1), the heads of machines with column diameters larger than 10 inches, shall be traversed by means of a power operated mechanism. When power traverse is provided, the head shall incorporate an interlock or slip clutch to prevent damage from engagement of the rapid traverse while the head is in a locked position. The spindle and all rotating members within the head shall be mounted on ball or roller bearings. A pump shall furnish a constant supply of oil to all sliding surfaces, rotating shafts or sleeves and bearings subject to frictional contact.

3.4.1.4.1 Spindle. The spindle shall be of heat-treated alloy steel with a surface hardness not less than Rockwell C-40. The driven portion of the spindle shall be multiple splined in accordance with ANS B17.2. The spindle shall be mounted in a sleeve which shall support or drive the spindle. The spindle shall be supported by or roller bearings. A spindle counterbalance mechanism shall be fully enclosed in the head. The spindle taper shall be of the size specified in table I, and shall comply with the requirements of ASA B5.10.

3.4.1.4.2 Speeds and feeds. The spindle speed and feed change controls shall be provided on the head. Speed changes shall be controlled by not more than three conveniently located levers and accomplished through sliding gears or a combination of sliding gears and positive tooth clutches. All speed and feed gears shall be of heat treated alloy steel. Sliding gears shall slide on multiple splined shafts having not less than three splines with integral keys. A depth gage shall be provided which shall automatically disengage the power feed at any predetermined depth indicated on the graduated dial. A device shall be included to automatically disengage the feed when the limit of spindle travel is reached. The feed engagement unit shall be operated by a positive tooth or friction type clutch with means for adjustment. When required, heads of machines with column diameters larger than 12 inches, shall have preselection of spindle speeds and feeds, as specified (see 6.2.1).

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TABLE I - Sizes and capacities - A4 through D2

Size	A4	A6	B1	B2	B3	C1	C2	D1	D2
Column diameter, inches	9	9	11	11	11	13	13	15	15
Arm length, feet	3	4	3	4	5	4	5	4	5
Spindle to base, maximum, inches	50	50	54	54	54	57	57	66	66
Spindle to base, minimum, not more than, inches	16	16	16	16	16	13	13	17	17
Arm bearing length on column, inches	18	18	22	22	22	26	26	30	30
Arm traverse on column, inches	26	26	25	25	25	29	29	35	35
Spindle traverse in head, inches	10	10	12	12	12	18	18	18	18
Head traverse on arm, inches	25	37	25	37	49	32	44	32	44
Spindle center to column, minimum, not more than, inches	13	13	11	11	11	16	16	16	16
Base height, inches	6	6	8	8	8	8	8	8	8
Base working length, inches	41	53	41	53	65	53	65	53	65
Base working width, inches	30	30	36	36	36	36	36	40	40
Spindle speeds, number of	9	9	12	12	12	32	32	32	32
Spindle speed range, RPM	90 to 1500	90 to 1500	60 to 1500	60 to 1500	60 to 1500	15 to 1200	15 to 1200	15 to 1200	15 to 1200
Spindle feeds, number of	4	4	6	6	6	16	16	16	16
Spindle feed range, inch per revolution	.004 to .020	.004 to .020	.006 to .025	.006 to .025	.006 to .025	.004 to .125	.004 to .125	.004 to .125	.004 to .125
Spindle diameter at point of drive, inches	1-3/8	1-3/8	1-1/2	1-1/2	1-1/2	2-1/2	2-1/2	2-3/16	2-3/16
Motor horsepower	5	5	5	5	5	10	10	15	15
Taper in spindle, Morse No	4	4	4	4	4	5	5	5	5
Tap leads, threads per inch: 1	-	-	-	-	-	8	8	8	8
2	-	-	-	-	-	11-1/2	11-1/2	11-1/2	11-1/2
3	-	-	-	-	-	14	14	14	14

Requirements are not less than the values shown. Where a range is shown, the requirement is from not more than the lower value to not less than the higher value.

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TABLE I - Sizes and capacities - D3 through G1

Size	D3	E1	E2	E3	F1	F2	F3	G1
Column diameter, inches	15	17	17	17	19	19	19	22
Arm length, feet	6	5	6	7	6	7	8	7
Spindle to base, maximum, inches	66	72	72	72	78	78	78	96
Spindle to base, *minimum, not more* than, inches	17	17	17	17	16	16	16	22
Arm bearing length on column, inches	30	34	34	34	38	38	38	44
Arm traverse on column, inches	35	37	37	37	43	43	43	50
Spindle traverse in head, inches	18	18	18	18	18	18	18	24
Head traverse on arm, inches	56	46	58	70	57	69	79	67
Spindle center to column, minimum, not more than, inches	16	15	15	15	15	15	15	17
Base height, inches	8	10	10	10	10	10	10	12
Base working length, *inches*	77	65	77	89	77	89	101	90
Base working width, inches	40	50	50	50	50	50	50	57
Spindle speeds, number of	32	32	32	32	32	32	32	36
Spindle speed range, RPM	15 to 1200	15 to 1200	15 to 1200	15 to 1200	15 to 1200	15 to 1200	15 to 1200	8 to 800
Spindle feeds, number of	16	16	16	16	16	16	16	18
Spindle feed range, inch per revolution	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125
Spindle diameter at point of drive, inches	2-3/16	2-1/2	2-1/2	2-1/2	2-1/2	2-1/2	2-1/2	3
Motor horsepower	15	15	15	15	20	20	20	20
Taper in spindle, Morse No.	5	5	5	5	6	6	6	6
Tap leads, threads per inch:								
1	8	8	8	8	8	8	8	8
2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2
3	14	14	14	14	14	14	14	14

Requirements are not less than the values shown. Where a range is shown, the requirement is from not more than the lower value to not less than the higher value.

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TABLE I - Sizes and capacities - G2 thrugh H5

Size	G2	G3	G4	H1	H2	H3	H4	H5
Column diameter, inches	22	22	22	26	26	26	26	26
Arm length, feet	8	9	10	8	9	10	11	12
Spindle to base, maximum, inches	96	96	96	100	100	100	100	100
Spindle to base, minimum, not more than, inches	22	22	22	22	22	22	22	22
Arm bearing length on column, inches	44	44	44	52	52	52	52	52
Arm traverse on column, inches	50	50	50	50	50	50	50	50
Spindle traverse in head, inches	24	24	24	24	24	24	24	24
Head traverse on arm, inches	79	91	103	80	92	104	116	128
Spindle center to column, minimum, not more than, inches	17	17	17	22	22	22	22	22
Base height, inches	12	12	12	12	12	12	12	12
Base working length, inches	102	114	126	102	114	126	138	150
Base working width, inches	57	57	57	57	57	57	57	57
Spindle speeds, number of	36	36	36	36	36	36	36	36
Spindle speed range, RPM	8 to 800	8 to 800	8 to 800	8 to 800	8 to 800	8 to 800	8 to 800	8 to 800
Spindle feeds, number of	18	18	18	18	18	18	18	18
Spindle feed range, inch per revolution	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125	.004 to .125
Spindle diameter at point of drive, inches	3	3	3	3	3	3	3	3
Motor horsepower	20	20	20	20	20	20	20	20
Taper in spindle, Morse No.	6	6	6	6	6	6	6	6
Tap leads, threads per inch:								
1	8	8	8	8	8	8	8	8
2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2	11-1/2
3	14	14	14	14	14	14	14	14

Requirements are not less than the values shown. Where a range is shown, the requirement is from not more than the lower value to not less than the higher value.

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3.4.1.5 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall be in accordance with JIC Electrical Standards for General Purpose Machine Tools or NFPA No. 70 or NFPA No. 79. A grounding terminal shall be provided for grounding the machine when installed. The terminal shall be suitable for connection of the size conductor specified in table 15-1 of JIC Standards. Except as indicated below, each machine shall operate on 230/460 volt, 3 phase, 60 hertz AC. Unless otherwise specified (see 6.2.1), initial wiring shall be for operation on 230 volts. When two-speed motors are required, the machine shall operate on the single voltage specified.

3.4.1.5.1 Motors. Motors shall be continuous duty and shall have ball or roller bearings. Unless otherwise specified (see 6.2.1), each motor shall be single speed and shall meet the requirements of a drip-proof enclosure.

3.4.2 Sizes and capacities. Unless otherwise specified (see 6.2.1), the machine shall meet the requirements of table I for the applicable machine size specified in the contract or order.

3.4.3 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. A coolant system to include a sump and strainer in the base, a power driven pump, piping and all electrical wiring needed for complete operation. The coolant system shall include oil resistant hose, tubing or piping to direct coolant and shall be fitted with a control valve.

b. One set of wrenches, operating and repair tools normally furnished by the manufacturer.

3.4.4 Optional equipment. The following optional equipment and accessories shall be as specified (see 6.2.1):

a. A tapping attachment unless a tapping feature is built-into the machine.

b. One worklight, adjustable to illuminate the work area, complete with protective shield and on-off switch.

3.4.5 Spare parts. Spare parts shall be furnished as specified (see 6.2.1).

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3.4.6 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461.

3.4.7 Fungus control. When required (see 6.2.1), fungus proofing shall be as specified.

3.4.8 Lubrication chart or plate. When specified (see 6.2.1), a lubrication chart or plate shall be securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or laminated between clear plastic, permanently sealed sheets with suitable means for mounting. The following information shall be furnished on charts or plates:

Points of application
Service interval
Type of lubricant
Viscosity
Federal or Military specification number (if available)

3.4.9 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate, permanently and legibly marked with the following information in accordance with MIL-STD-130, shall be securely attached to each drilling machine:

Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Electrical input (volts, total amps, phase, frequently)
Contract Number
Federal Stock Number, National Stock Number or Plant Equipment Code
Date of manufacture
US

3.4.10 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.4.11 Workmanship. The drilling machine and its accessories shall be free from defects such as incomplete welds, rust, cracks, or any other defect that could impair its operation or serviceability.

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3.5 Performance. The machine shall be capable of performing drilling, tapping, boring, and reaming operations for accomplishing all functional characteristics for which it is designed along with those required by table I. The machine shall also be capable of performing full horsepower drilling without any evidence of chatter, stalling, or permanent distortion.

3.5.1 Alignment and accuracy. The machine shall meet the alignment and accuracy tolerances required by table II.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article inspection. When first article approval is required under 3.2, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure to pass the examination or any test shall be cause for disapproval of the first article.

4.3 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4 and 4.6, and the tests in 4.5. Failure of the machine to pass the examination or any test shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined for compliance with the requirements herein in 3.3 through

4.5 Test procedures.

4.5.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms and other accessories shall be inspected during the trial period.

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4.5.2 Horsepower test. The machine shall be subjected to a drilling test which applies not less than half the rated input power to the spindle motor for at least one minute. Spindle speed shall be chosen by the supplier at the lowest practicable for this test without applying torque in excess of the design maximum value to any machine components. Material and dimensions of the workpiece, tooling, depth of cut, and feed rate shall be at the option of the supplier to achieve loading. There shall be no evidence of chatter, stalling, or permanent distortion during or after the test.

4.5.3 Alignment and accuracy test. The drilling machine and its components shall be tested for conformance with the alignment and accuracies specified in table II.

4.5.3.1 Test data sheets. A copy of the test sheets covering 4.5.3 shall be furnished with the machine.

4.5.4 Electromagnetic interference testing. Equipment which requires electromagnetic interference control testing should be required to be tested in accordance with MIL-STD-461.

4.6 Examination of preparation for delivery. Examination of preparation for delivery shall be accomplished to determine compliance with the requirements of Section 5.

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TABLE II. Alignment and Accuracy Tolerances.

COMPONENTS	MAXIMUM ALLOWANCE DEVIATION
Base	
Work surface, flat in longitudinal direction	0.0012 inch per foot
Work surface, flat in traverse direction	0.0012 inch per foot
Column[1]	
Column square with base in plane through center line of base	
For machines up to 4 foot 7 inch length radial arm	0 to 0.0024 inch per foot
For machines over 4 foot 7 inch length of radial arm	0 to 0.0036 inch per foot
Column square with base in plane perpendicular to plane through center line of baseplate	0.0012 inch per foot
Arm[1]	
Arm parallel with base (low at outer end)	
For machines up to 4 foot 7 inch length of radial arm	0 to 0.0024 inch per foot
For machines over 4 foot 7 inch length of radial arm	0 to 0.0036 inch per foot
Arm ways, flat	0.0012 inch per foot
Drilling spindle	
Spindle runout (using 12 inch long mandrel)	0.002 inch
Spindle square with base in plane through center line of base (lower end inclined toward column)	0 to 0.0024 inch per foot
Spindle square with work surface of base in plane 90 degrees to above plane.	0.0012 inch per foot
Travel of spindle sleeve square with work surface of base in plane through center line of base (inclined at lower end toward column)	0 to 0.004 inch per foot
Travel of spindle square with base in plane 90 degrees to above plane	0.002 inch per foot
[2]Drill spindle square with plain type table mounted on base in plane through center line of base (spindle inclined at lower end toward column)	0 to 0.0024 inch per foot
[2]Drill spindle square with plain type table mounted on base in plane perpendicular to above plane	0.0012 inch per foot

[1]When testing, the arm shall be positioned half-way up the column and the head half-way along the arm.
[2]When a plain type table is furnished.

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5. PREPARATION FOR DELIVERY

5.1 Military agencies. Preservation, packing, and marking shall be in accordance with the applicable requirements or MIL-M-18058. The required level of preservation and packing, and any special marking requirements shall be as specified (see 6.2.1).

5.2 Civil agencies. Preservation, packing, and marking shall be in accordance with the applicable requirements of FED. STD. No. 356. Special marking requirements shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. The radial drilling machines covered by the specification are intended for use in drilling, reaming, spot facing, tapping, boring and counterboring operations on work too large or too heavy to be placed on an upright drilling machine.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

(a) Title, number and date of this specification.
(b) Size of machine required (see 1.2.1).
(c) First Article approval, if required (see 3.2 and 6.3).
(d) Specify exceptions, and additional requirements for safety and health (see 3.3.1.1 and 6.5).
(e) Painting, if different (see 3.3.2.5).
(f) Threaded parts, if different (see 3.3.2.6)
(g) Dials, if different (see 3.3.2.9).
(h) Plain type table, if required (see 3.4.1.1.1).
(i) Universal tilting type table, if required (see 3.4.1.1.2).
(j) Clamping devices for machines with column diameters larger than 13 inches, if required (see 3.4.1.3).
(k) Power traverse for machines with column diameters larger than 10 inches, if required (see 3.4.1.4).
(l) Specify degree of preselection of spindle speeds and feeds, if required for machines with column diameters larger, than 12 inches, (see 3.4.1.4.2).

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(m) Electrical system, if different, and voltage for initial wiring, if different (see 3.4.1.5).
(n) Motors, if different (see 3.4.1.5.1).
(o) Sizes and capacities, if different, and voltage for initial wiring, if different (see 3.4.1.5).
(p) Standard equipment, if different (see 3.4.3).
(q) Optional equipment, if required (see 3.4.4).
(r) List spare parts required (see 3.4.5).
(s) Electromagnetic interference control, if required (see 3.4.6).
(t) Type of fungus proofing, if required (see 3.4.7).
(u) Lubrication chart or plate, if required (see 3.4.8).
(v) Nameplate, if different (see 3.4.9).
(w) Preproduction inspection, if different (see 4.2).
(x) Quality conformance inspection, if different (see 4.3).
(y) Level of preservation-packing and special marking required (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as operator's manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. A first article comprises a preproduction item or standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examination, test and approval of the first article.

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6.5 Safety and health environmental determinations. In order that equipment integrated into the user's operational environment will comply with the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc. as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

6.6 Size relation to previous revision. Sizes in this specification are identical with the sizes in the previous revision.

6.7 Mercury prohibited. When the use of mercury or its compounds in the machine and its components and accessories is to be prohibited, procurement documents should include a statement to that effect.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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